

A dynamic rock–TBM spatial relation graph for component-level geological anomaly interpretation during TBM shield jamming

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Abstract

Mechanized tunnelling creates a moving spatial configuration in which weak or anomalous ground becomes relevant to the tunnel boring machine (TBM) through changing relations with specific machine components rather than through a machine-wide contact zone. TBM monitoring records describe operational responses such as shield pressure, displacement, thrust and advance rate, while tunnel seismic prediction (TSP) data describe the spatial support of geological anomalies. These data sources do not, by themselves, identify which geological units are spatially related to the cutterhead, front shield, middle shield or tail shield at a given excavation step. This paper proposes a dynamic rock–TBM spatial relation graph that represents TSP-derived rock voxels and component-labelled TBM surface nodes as heterogeneous spatial entities and retains screened candidate rock–TBM relations as edge-level spatial supports. Candidate relations are generated under active-zone, distance, surface-normal compatibility and excavation-state constraints, and edge-weighted geological anomaly scores are aggregated into a component-level geological anomaly index. The evaluation is framed as event-centred representation validation using a confirmed TBM shield-jamming event in the Boshulaling Tunnel. It examines relation reconstruction, component specificity, comparisons with alternative geological summaries, sensitivity to graph-construction settings and edge-level provenance. The results show that the graph converts TSP-derived low-velocity anomalies into component-indexed relation evidence, distinguishes shield-related anomaly support from cutterhead or pooled TBM summaries, and allows the interpretation to be traced back to individual rock voxels, TBM surface nodes and candidate edges. The method is therefore intended as a queryable spatial relation representation for component-level TBM anomaly interpretation, rather than as a calibrated failure-prediction or contact-force estimation model.

Keywords: dynamic spatial relation graph; rock–TBM relation; component-level geological anomaly index; TBM shield jamming; edge-level provenance; TSP-derived rock voxels

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1 Introduction

TBM excavation is a moving spatial relation problem. As the cutterhead and shield advance, geological units enter, approach and leave the potential relation zone of different TBM components. Changes in shield pressure, displacement, thrust, torque, penetration and advance rate therefore reflect not only operational control but also the evolving spatial configuration between geological anomalies and machine geometry. The GIScience problem addressed here is the representation of these changing rock–machine relations under heterogeneous spatial supports.

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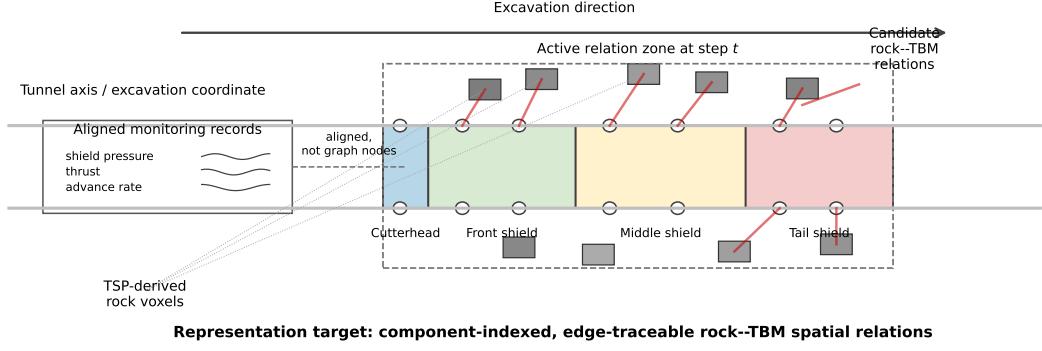


Figure 1: Research scene and object of representation. A TSP-derived geological voxel field and a component-labelled advancing TBM surface are observed under different spatial supports. The proposed representation constructs candidate rock–TBM relations between anomalous rock voxels and machine components, while monitoring responses are aligned to the resulting component-level indices for event interpretation.

In this paper, the engineering event is referred to as *TBM shield jamming*. The term denotes a jammed state in which surrounding ground convergence and shield-side resistance make continued advance difficult or impossible under the available thrust. The term is preferred to “shield sticking” because the tunnelling literature more commonly uses TBM jamming or shield jamming when describing squeezing-ground obstruction, shield skin-friction resistance, and jamming accidents. “Stuck” or “jammed” is used only as a state descriptor, not as the main technical term.

Monitoring-based TBM response models are useful because short-horizon excavation responses often have strong temporal persistence. Empirical, machine-learning and sequence-based models can learn response inertia from rock, machine and recent operational histories [2, 11, 12, 32]. However, a monitoring curve does not state which geological voxel is adjacent to the cutterhead or shield, whether that adjacency is geometrically plausible, or how the relation support changes as excavation proceeds. This limits the spatial interpretability of monitoring-only analyses.

Advance geological prediction and TSP surveys provide a complementary view by describing spatial heterogeneity ahead of or around the excavation face [16]. These data can be represented as chainage-referenced geological voxels or attribute fields, preserving where low-velocity or other anomalous ground occurs. Yet a geological voxel field answers where an anomaly is located, not which TBM component is a plausible relation partner at a particular excavation step. The missing object is therefore neither another monitoring feature nor a single geological average, but a component-indexed relation support between geological spatial units and machine-side spatial units.

A moving buffer or a linear-referenced overlay is insufficient for this task. A buffer can summarise the mean anomaly inside a moving active zone, but it does not retain which rock voxel is related to which TBM component, which edge attributes make the relation plausible, or whether the relation appears, persists or disappears over excavation steps. The proposed graph keeps heterogeneous entities, component labels, candidate-edge attributes and edge lifecycle explicit. The resulting component–excavation-step index is a relation aggregation over screened candidate edges rather than a simple voxel-field average.

This paper therefore treats rock–TBM interpretation as a spatial entity–relation modelling problem. The input consists of a TSP-derived rock voxel field, an advancing parameterized TBM

surface and excavation-step-aligned monitoring records. Two graph entity types are defined: rock voxels and TBM surface nodes. Monitoring responses are aligned to graph outputs but are not used to define graph nodes or rock–TBM edges. The method constructs a dynamic rock–TBM spatial relation graph at each excavation step using active-zone, distance, surface-normal compatibility and excavation-state constraints. Weighted candidate edges are aggregated by TBM component to compute a component-level geological anomaly index.

The paper makes three contributions. First, it defines a rock–TBM spatial entity model that places TSP-derived rock voxels and component-labelled TBM surface nodes in a common excavation-step coordinate while keeping monitoring responses as aligned records. Second, it constructs a sparse per-step candidate relation graph using physically interpretable spatial constraints rather than full connectivity or unconstrained learned edges. Third, it computes and evaluates a component-level geological anomaly index that supports relation reconstruction, component specificity, comparison with alternative summaries, sensitivity checks and edge-level provenance. The empirical evidence is used to validate the spatial relation representation in a confirmed TBM shield-jamming event.

2 Related work

2.1 TBM jamming and monitoring-based response modelling

TBM jamming is a recurrent problem in difficult or squeezing ground because the surrounding rock may converge toward the machine, increase shield loading and generate frictional resistance along the shield. The literature usually distinguishes machine-area problems such as cutterhead sticking and shield jamming, and more recent data-driven studies also use the broader expression TBM jamming accidents. In the present paper, the case is described as TBM shield jamming because the abnormal response is dominated by shield-side pressure, shield displacement, high thrust demand and loss of advance rather than by a purely cutterhead-local obstruction.

Traditional TBM performance studies estimate penetration, advance rate, thrust, torque or related operational responses from rock mass properties and machine parameters [2, 11, 32]. With the accumulation of monitoring data, sequence models also learn lagged dependence from recent operational histories [4, 12]. These studies provide important response-prediction baselines, but their spatial unit is usually a tunnel section, a sample or a feature vector. A monitoring curve can describe how thrust, torque, penetration, advance rate or shield pressure changes with time or chainage, but it does not identify which geological voxels are plausible relation partners for the cutterhead, front shield, middle shield or tail shield. The present study therefore does not replace monitoring-based predictors. It asks whether a spatial relation representation provides component-indexed information about the rock–machine configuration that monitoring curves alone do not encode.

2.2 Geological prediction and the component-relation gap

Advance geological prediction methods, including TSP and related geophysical or drilling-based approaches, are used to identify faults, fractured zones, water-rich zones, weak surrounding rock and other geological anomalies before or during tunnel excavation [16]. For TBM interpretation, such data are valuable because they describe spatial heterogeneity that is not contained in the monitoring record itself. Voxelisation is useful because it preserves spatial support, local attribute values and chainage-referenced geological context.

Geological spatial representation alone is nevertheless incomplete for component-level interpretation. A voxel field can show where anomalous ground occurs, but it does not specify whether that anomaly is spatially related to the cutterhead or shield, whether the relation is surface-normal compatible, or whether the voxel remains active at the current excavation step. TSP voxels, TBM surface samples and monitoring records have different spatial or observational

supports and should not be collapsed into a single tabular unit without retaining relation information [10, 19, 20]. The missing representation is a relation layer that links geological spatial supports to component-labelled machine-side supports during excavation.

2.3 Spatial relation graphs for moving excavation scenes

GIScience is concerned not only with locations but also with geographical objects, spatial relations, temporal change and processes [8, 10, 27]. Spatial relation theory provides concepts for neighbourhood, adjacency, containment, intersection, distance and orientation [6, 28]. TBM excavation fits this perspective because rock units, machine surfaces and monitoring responses are observed in a moving excavation coordinate, and the relevant relations change as the machine advances.

Recent geospatial knowledge-graph and graph data-model studies show how heterogeneous spatial entities can be integrated for retrieval, reasoning, query and provenance [15, 22, 31, 37]. Scenario-oriented spatiotemporal knowledge-graph models likewise emphasise linking heterogeneous data through meaningful objects and relations [25]. The present paper shares this entity–relation modelling perspective, but it operates at a finer voxel and surface-node support and targets a moving underground engineering object rather than an urban knowledge service or indoor social network. Rock voxels are geological spatial objects, TBM surface nodes are machine-side spatial objects, and rock–TBM candidate edges encode screened spatial relation support.

Graphs are also a natural formalism for representing irregular spatial objects and their relations. Recent geospatial graph studies emphasise that edges should be related to the spatial process of interest and should not be treated as arbitrary connections [3, 5, 18, 36]. Spatial interaction models further emphasise that relations are shaped by distance decay, compatibility, direction, accessibility and spatial support [7, 26]. For the rock–TBM problem, candidate relations should depend on active-zone membership, distance decay, surface-normal compatibility and excavation state. This is different from a fully connected graph or a purely learned adjacency matrix. Graph neural networks, heterogeneous graph models and spatiotemporal graph models remain relevant as future extensions for learning on structured data [9, 17, 21, 23, 24, 29, 30, 33–35]. Digital-twin frameworks in tunnelling likewise motivate structured links among monitoring, prediction and interpretation modules [1, 13, 14]. The present study adopts the evaluation spirit of recent spatiotemporal graph studies—comparison with static, pooled or ablated alternatives—but applies it to a spatial representation problem. The target is not higher forecasting accuracy, but evidence that the proposed graph provides relation reconstruction, component specificity, robustness to graph-construction settings and edge-level provenance beyond simpler geological summaries.

3 Methodology

This section formalises rock–TBM interpretation as an excavation-step-indexed spatial entity–relation graph. The method starts from a TSP-derived rock voxel field, an advancing parameterized TBM surface and aligned monitoring records under a common excavation-step index. It then constructs a sparse dynamic rock–TBM spatial relation graph at each excavation step and aggregates edge-weighted geological anomaly scores into a component-level geological anomaly index. The graph is sparse because candidate rock–TBM relations are retained only when the four spatial and excavation-state constraints are jointly satisfied.

Given geological voxels D_{geo} , a parameterized TBM surface M_{TBM} and excavation step t , the graph-construction operator is

$$G : (D_{geo}, M_{TBM}, t) \mapsto G_t. \quad (1)$$

Table 1: Entities, relations and indices in the revised rock-TBM spatial relation graph.

Entity or quantity	Definition	Role
V_t^r	Active TSP-derived rock voxels	Geological spatial support
V_t^m	Component-labelled TBM surface nodes	Machine-side spatial support
E_t^{rr}	Rock-rock voxel neighbourhoods	Geological contiguity context
E_t^{mm}	TBM surface neighbourhoods	Machine-surface topology context
E_t^{rm}	Screened rock-TBM candidate relations	Candidate relation support
$A_c(t)$	Sum of geometry weights for component c	Geometric exposure metric
$I_c(t)$	Edge-weighted anomaly intensity for component c	Main component-level index
$e_{t+h}^{(k)}$	Persistence residual of response k	Aligned monitoring residual

The field-computation operator is

$$F : (G_t, \phi(g_i), \rho_j) \mapsto \{I_c(t), \Delta I_c(t)\}_{c \in C, t=1, \dots, T}, \quad (2)$$

where $\phi(g_i)$ maps geological attributes to a rock-voxel anomaly score and ρ_j is the TBM component label. Monitoring records are used to read the resulting index against excavation responses; they are not inputs to G and do not define graph edges.

3.1 Rock, machine and monitoring data representation

The input data are organised into two graph entity types and aligned monitoring records: rock voxels, TBM surface nodes and excavation monitoring responses. Monitoring records are not graph nodes; they are used after graph computation to evaluate and interpret the component-indexed geological anomaly index.

Rock voxel. The geological input is a TSP-derived voxel field $D_{geo} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector, such as P-wave velocity V_p , S-wave velocity V_s , dynamic Young’s modulus, velocity ratio, Poisson’s ratio or density. Rock voxels are the geological spatial support of the graph.

TBM surface node. The machine input is a parameterized TBM surface. Each surface node is represented by $(p_j(0), n_j, \rho_j)$, where $p_j(0)$ is the initial surface-node coordinate, n_j is the local outward surface normal and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield and tail shield components.

Aligned monitoring records. At each excavation step t , the monitoring vector is

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t]. \quad (3)$$

The response variables used downstream are denoted $r_t^{(k)}$, where k indexes a response channel. Monitoring responses are aligned to $I_c(t)$ and $\Delta I_c(t)$; they do not participate in graph construction.

Common chainage coordinate. TSP coordinates, TBM geometry and monitoring records are mapped to a common local excavation coordinate. With $\text{ShieldMileage}(t)$ the shield mileage stamp at step t ,

$$x_{local}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{start}, \quad (4)$$

and TBM surface nodes are advanced along the excavation direction d :

$$p_j(t+1) = p_j(t) + \Delta x d. \quad (5)$$

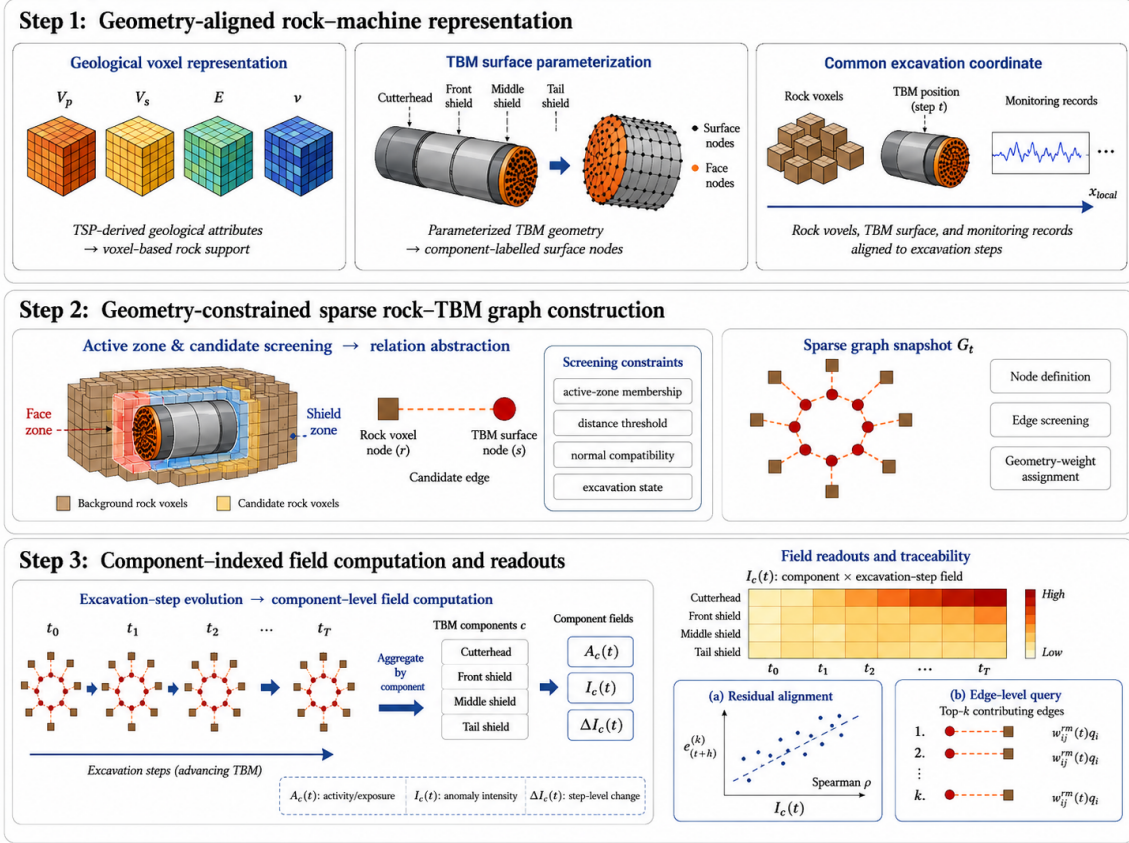


Figure 2: Workflow of the dynamic rock-TBM spatial relation graph. Rock voxels, TBM surface nodes and aligned monitoring records are organised under a common excavation-step coordinate. Candidate rock-TBM relations are generated under active-zone, distance, surface-normal compatibility and excavation-state constraints, aggregated into component-level geological anomaly indices, and queried back to edge-level provenance.

3.2 Geometry-constrained sparse graph construction

At each excavation step, a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}). \quad (6)$$

Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)], \quad (7)$$

where $s_i(t)$ records whether voxel i remains active and eligible for relation construction at step t . TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j]. \quad (8)$$

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The central design decision is the construction of E_t^{rm} , the rock-TBM candidate relation set.

A candidate edge (i, j) is included in E_t^{rm} only when four conditions are jointly satisfied:

$$\begin{cases} c_i \in \Omega_t, \\ d_{ij}(t) \leq \tau_{edge}, \\ \kappa_{ij}(t) \geq \eta_{min}, \\ s_i(t) = 1. \end{cases} \quad (9)$$

Placeholder for candidate-relation construction: active-zone, distance,
normal-compatibility and excavation-state filtering.

Figure 3: Geometry-constrained rock–TBM candidate relation construction. The graph is obtained by screening all possible rock voxel–TBM surface-node pairs through active-zone membership, distance, surface-normal compatibility and excavation-state constraints.

The first rule restricts relations to the excavation-relevant active rock volume Ω_t ; the second uses

$$d_{ij}(t) = \|c_i - p_j(t)\| \quad (10)$$

to retain spatially close rock–machine pairs; the third uses the signed surface-normal compatibility

$$\kappa_{ij}(t) = \max \left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon} \right) \quad (11)$$

to retain geometrically plausible facing relations; and the fourth prevents already excavated voxels from repeatedly contributing to later graph snapshots.

The active zone separates face-contact plausibility from shield-side proximity:

$$\Omega_t = \Omega_t^{face} \cup \Omega_t^{shield}, \quad (12)$$

with

$$\Omega_t^{face} = \{i \mid 0 \leq x_i - x_f(t) \leq L_f, r_i \leq R_c + \tau_{zone}\}, \quad (13)$$

$$\Omega_t^{shield} = \{i \mid x_{tail}(t) \leq x_i \leq x_f(t), R_s \leq r_i \leq R_s + \tau_{zone}\}. \quad (14)$$

For each retained candidate relation, the geometry-compatibility weight is

$$w_{ij}^{rm}(t) = \exp \left(-\frac{d_{ij}(t)}{\tau_{edge}} \right) \kappa_{ij}(t). \quad (15)$$

This quantity is a spatial relation weight; it is not a contact force, pressure or friction estimate.

3.3 Component-level geological anomaly index

The geological contribution of each rock voxel is represented by a bounded TSP-derived anomaly score

$$q_i = \phi(g_i). \quad (16)$$

In the main formulation, q_i is a low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip} \left(\frac{Q_{95}^{train}(V_p) - V_{p,i}}{Q_{95}^{train}(V_p) - Q_5^{train}(V_p) + \epsilon}, 0, 1 \right). \quad (17)$$

This fixed training-reference scaling avoids per-chainage min–max normalisation and keeps values comparable along chainage; lower V_p values correspond to higher anomaly scores under the common reference distribution.

For each TBM component c , let V_c^m denote the TBM surface nodes with component label c . The geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t), \quad (18)$$

and the component-level geological anomaly index is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{A_c(t) + \epsilon}. \quad (19)$$

The index is computed for the cutterhead, front shield, middle shield and tail shield at each excavation step.

Because the confirmed case is shield jamming, shield components are also summarised as an exposure-weighted shield-group index:

$$I_{shield}^A(t) = \frac{\sum_{c \in C_{shield}} A_c(t) I_c(t)}{\sum_{c \in C_{shield}} A_c(t) + \epsilon}, \quad C_{shield} = \{\text{front, middle, tail}\}. \quad (20)$$

This replaces the unweighted arithmetic mean and avoids assigning equal influence to components with different relation supports. A shield-specificity contrast is further defined as

$$S_{shield}(t) = \frac{\sum_{c \in C_{shield}} A_c(t) I_c(t)}{A_{cutterhead}(t) I_{cutterhead}(t) + \epsilon}. \quad (21)$$

The step-level change is

$$\Delta I_c(t) = I_c(t) - I_c(t-1). \quad (22)$$

For response alignment, short-horizon TBM responses are evaluated through persistence residuals rather than raw responses:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)}. \quad (23)$$

Association statistics between $I_c(t)$ and $e_{t+h}^{(k)}$ are interpreted as exploratory index-residual co-variation within the evaluated chainage interval, not as causal attribution.

For edge-level provenance, candidate edges for any diagnostic sample are ranked by

$$s_{ij}(t) = w_{ij}^{rm}(t) q_i. \quad (24)$$

Each record retains rock voxel identity, TBM surface-node identity, component label ρ_j , distance $d_{ij}(t)$, normal compatibility $\kappa_{ij}(t)$, geometry weight $w_{ij}^{rm}(t)$, anomaly score q_i and weighted contribution $s_{ij}(t)$.

4 Case data and representation-validation design

4.1 Shield-jamming event and aligned data

The evaluation uses a confirmed shield-jamming event in the right line of the Boshulaling Tunnel. The event was confirmed at 16:30 on 3 August 2024 after increasing shield pressure, converging shield displacement, increasing total thrust and nearly zero advance rate while the cutterhead continued rotating. The event occurred near chainage DyK1017+205, approximately 72 m before the mapped F37 fault influence zone. The engineering record therefore provides a shield-dominated abnormal episode rather than dense labels for every geological voxel or every TBM component.

This case is used as an event-centred representation-validation test. The objective is not to train a general failure-prediction model, estimate ground-machine contact force or derive a calibrated risk probability. Instead, the evaluation asks whether the graph connects three forms of evidence in a common excavation coordinate: TSP-derived geological anomalies, component-labelled TBM geometry and monitoring responses during the shield-jamming process.

Table 2: Case event, input data and alignment used for representation validation.

Data group	Main fields	Role in the evaluation
Event log	Confirmed jamming time, event chainage, event-development window, engineering description	Defines the shield-jamming event and the event window
TSP geological data	Rock voxel coordinates, V_p , V_s where available, geological anomaly score q_i	Provides rock voxels and TSP-derived geological anomaly scores
TBM monitoring	Shield pressure, shield displacement, total thrust, advance rate, cutterhead RPM, timestamp and chainage	Describes machine response during the event
TBM geometry	Cutterhead, front shield, middle shield and tail shield surface nodes	Provides component-labelled TBM surface nodes
Alignment data	Timestamp-chainage mapping, TSP local coordinate and event chainage	Links geological field, monitoring records and graph snapshots

4.2 Representation-validation protocol

The event-centred evaluation is organised around five tasks. E1 tests whether the graph re-constructs a spatial relation process in which low-velocity rock voxels become associated with shield components as the TBM approaches the confirmed jamming location. E2 tests whether the component-level index carries information that is lost in global or pooled summaries. E3 tests whether the relation pattern remains stable under reasonable graph-construction settings. E4 tests whether the interpretation can be traced back to individual rock voxels, TBM surface nodes and candidate rock-TBM edges. E5 reports exploratory response alignment using persistence residuals, without converting the result into a prediction benchmark.

4.3 Alternative representations and ablation settings

To avoid treating the event as a post-hoc visual explanation only, the proposed full graph is compared with four alternatives: (i) a global active-zone geological anomaly average, (ii) a pooled TBM anomaly summary without component labels, (iii) a component-level graph without surface-normal compatibility, and (iv) a component-level graph without the excavation-state constraint. These alternatives test whether the proposed representation adds value by retaining component labels, geometry-compatible candidate edges and edge lifecycle.

4.4 Sensitivity settings

Sensitivity checks vary the main graph-construction parameters around the formal setting. The formal run uses $\tau_{edge} = 2.0$ m, $\eta_{min} = 0.3$ and $\tau_{zone} = 5.0$ m. The evaluation reports whether the event-window shield-group index, shield-specificity contrast and top-edge provenance remain stable under neighbouring settings.

5 Results

5.1 Spatial setting of the shield-jamming event

Figure 4 places the shield-jamming event in its geological and excavation context. The confirmed event location lies approximately 72 m before the F37 fault influence zone, while the TSP-derived

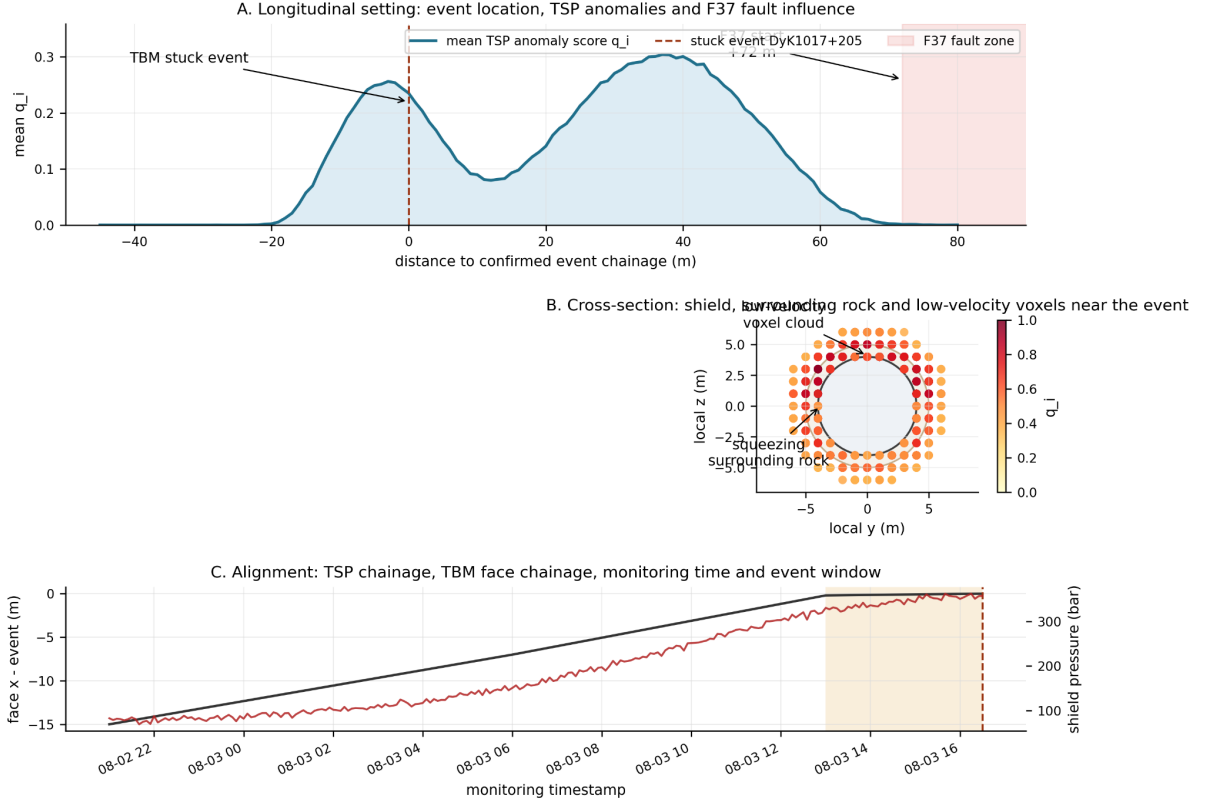


Figure 4: Spatial setting and data alignment for the shield-jamming event. Panel A shows the longitudinal event location, TSP-derived anomaly scores and the F37 fault influence zone. Panel B shows the local cross-section of the shield, surrounding rock and low-velocity voxels. Panel C aligns TBM face chainage, monitoring time and the event window.

geological anomaly score indicates weak ground both near the jamming location and further ahead toward the fault influence zone. The local cross-section shows that low-velocity voxels are distributed around the shield boundary, indicating that the event cannot be interpreted only from a one-dimensional geological profile. The aligned time-chainage panel further shows that the TBM face approached the confirmed event chainage during the monitoring period while shield pressure increased toward the event window.

This setting establishes the spatial representation problem addressed by the graph. TSP data indicate anomalous ground, and monitoring data indicate shield jamming, but neither directly represents which rock voxels are spatially related to which TBM components during excavation.

5.2 Dynamic rock-TBM relation-support evolution

Figure 5 shows three graph snapshots before and during the shield-jamming process. Before event development, the TBM is still separated from the strongest shield-related anomaly zone, and only a limited number of candidate rock-TBM relations are formed. At the beginning of the near-jamming window, more low-velocity voxels become connected to the front and middle shield components. Near the confirmed jammed state, candidate relations concentrate around the shield components, indicating that anomalous rock voxels are spatially associated with the shield rather than only with the cutterhead or the TBM as a whole.

Table 3: Representation-validation tasks used in the event-centred evaluation.

Task	Question	Main output	Claim supported
E1	Do candidate relations reconstruct a moving rock-shield relation process?	Dynamic graph snapshots; component edge shares; relation-support trajectories	Relation reconstruction
E2	Does the component-level index add information beyond simpler summaries?	Comparison against global active-zone anomaly, pooled TBM anomaly and ablated component summaries	Component specificity
E3	Is the pattern robust to graph-construction settings?	Sensitivity over τ_{edge} , η_{min} and τ_{zone}	Parameter robustness
E4	Can the interpretation be traced back to edges?	Top- k edge-provenance table and graph view	Edge-level provenance
E5	Is the index aligned with response changes in the event interval?	Exploratory Spearman association with persistence residuals	Response-aligned interpretation

5.3 Component-indexed anomaly evolution

Figure 6 summarises the same process in a component-chainage form. The upper panel shows the TSP anomaly band along the TBM face distance to the confirmed event chainage. The middle panel shows the component-level geological anomaly index for the cutterhead, front shield, middle shield and tail shield. The shield components exhibit increasing anomaly support as the TBM approaches the confirmed jamming location. The lower monitoring strips show the corresponding machine response: shield pressure and thrust increase, shield displacement converges, and the advance rate drops during the near-jamming window while cutterhead rotation remains non-zero.

The comparison is not interpreted as deterministic prediction. Instead, it shows spatial consistency among anomalous geological voxels, shield-related component-level indices and the observed shield-jamming response. This supports the event-centred claim that the graph represents an increase in candidate spatial relations between anomalous rock voxels and shield components.

5.4 Comparison with alternative summaries

Table 6 summarises the representational capacity of the proposed graph against simpler alternatives. A global active-zone anomaly summary can indicate that weak ground exists near the tunnel, but it cannot identify which TBM component is spatially related to the anomaly. A pooled TBM anomaly summary can indicate that the machine is near anomalous ground, but it removes the distinction among cutterhead, front shield, middle shield and tail shield. The no-normal and no-state ablations retain some component information but either admit geometrically implausible edge directions or allow already excavated voxels to re-enter later snapshots. The full graph preserves component identity, relation screening, edge lifecycle and edge-level provenance.

Table 4: Alternative summaries and ablation settings for representation validation.

Representation		What is retained	What it tests
Global anomaly	active-zone	Mean q_i within Ω_t	Whether a moving geological buffer is sufficient
Pooled anomaly	TBM	Edge-weighted anomaly over all TBM surface nodes	Whether component labels are necessary
No-normal component graph	compo-	Component labels and distance screening only	Whether surface-normal compatibility matters
No-state component graph	component	Component labels, distance and normal compatibility, but no excavation-state filtering	Whether edge lifecycle matters
Full proposed graph		Component labels, active zone, distance, normal compatibility and excavation state	Complete relation-support representation

Table 5: Recommended sensitivity checks for graph-construction parameters.

Parameter	Values	Stability output
τ_{edge}	1.0, 1.5, 2.0, 2.5, 3.0 m	Event-window $I_{shield}^A(t)$ peak and edge count
η_{min}	0.1, 0.3, 0.5	Shield/cutterhead contrast and candidate-edge sparsity
τ_{zone}	3.0, 5.0, 7.0 m	Active-zone anomaly support and top-edge provenance overlap

5.5 Sensitivity and null diagnostics

The sensitivity analysis should be reported as a compact robustness panel rather than as a prediction table. The expected result is not that every value is identical, but that the event-window shield-group index and shield-specificity contrast remain in the same chainage neighbourhood and that the top contributing edge set remains substantially overlapping under reasonable parameter changes. A useful stability measure is the top- k provenance overlap:

$$O_k(\theta, \theta_0) = \frac{|\mathcal{E}_k(\theta) \cap \mathcal{E}_k(\theta_0)|}{k}, \quad (25)$$

where θ_0 is the formal setting and $\mathcal{E}_k(\theta)$ is the top- k candidate-edge set under parameter setting θ .

Exploratory index-residual associations may be retained, but they should be presented as supplementary response-alignment diagnostics. They should not be used as the primary evidence for the method because the paper’s main claim is a spatial relation representation claim rather than a calibrated prediction claim.

5.6 Edge-level provenance

Figure 8 and Table 7 provide edge-level provenance near the jammed state. The local graph view shows low-velocity voxels, shield surface nodes and top contributing candidate rock–TBM edges. The provenance table records the component label, relative chainage, distance, normal compatibility, geological anomaly score, spatial relation weight and edge contribution for the top candidate edges.

This result shows that the component-level geological anomaly index is not an opaque aggregate. It can be traced back to specific anomalous rock voxels, shield surface nodes and candidate

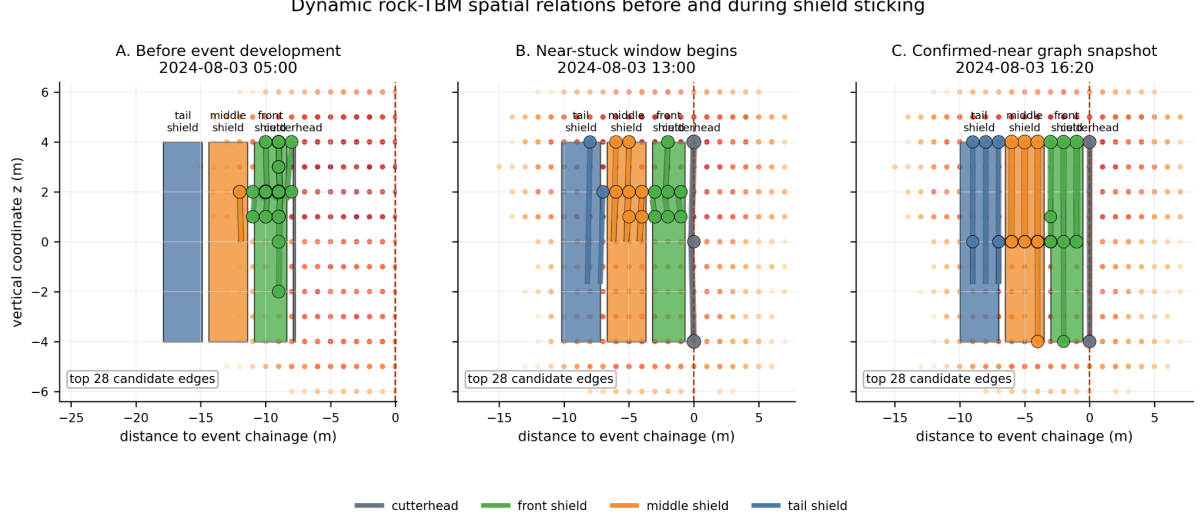


Figure 5: Dynamic rock–TBM spatial relation graph snapshots before and during shield jamming. Candidate edges are displayed with opacity proportional to $w_{ij}^{rm}(t)q_i$, and the side bar summarises edge contribution by TBM component.

spatial relations. This traceability is the key advantage of representing the TBM excavation process as a dynamic rock–TBM spatial relation graph.

6 Discussion

6.1 Why a relation graph rather than a moving overlay?

The proposed graph adds a relation layer between two commonly separated data sources. TSP data describe anomalous geological conditions, but they do not specify which TBM components are geometrically related to the anomalous voxels during excavation. TBM monitoring records describe machine responses, but they do not identify the spatial source of the response in the surrounding geological field. The graph connects these perspectives by representing candidate spatial relations between rock voxels and component-labelled TBM surface nodes.

This contribution is different from producing another monitoring indicator. The component-level geological anomaly index is derived from candidate rock–TBM relations and therefore preserves component identity, relation-screening attributes, edge lifecycle and edge provenance. These properties are lost in global geological summaries or pooled TBM summaries. They are essential for shield jamming because the abnormal response is component-specific rather than a machine-wide phenomenon.

6.2 What the event evidence supports

The event-centred evidence supports four representation claims. First, the graph reconstructs a moving relation-support process in which anomalous rock voxels become increasingly associated with shield components as the TBM approaches the confirmed event. Second, the component-level index provides more specific evidence than global or pooled geological summaries. Third, the relation pattern can be checked against parameter sensitivity and ablation settings. Fourth, the interpretation can be traced back to specific candidate edges and their geological and geometric attributes.

The evidence should be read at this representation level. The spatial relation weight is a geometry-compatibility weight, not a measurement of ground–machine contact force, shield

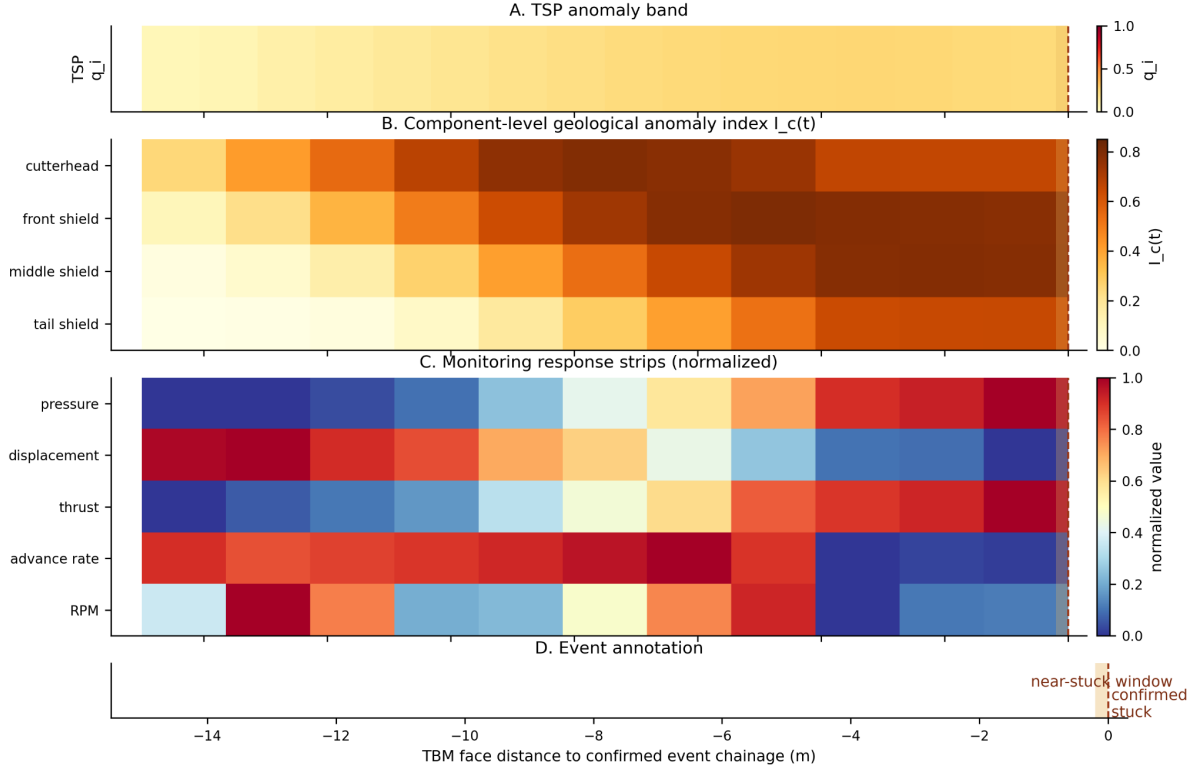


Figure 6: Component-indexed geological anomaly evolution and monitoring responses. The figure aligns the TSP anomaly band, component-level geological anomaly indices, shield-specificity contrast and normalized monitoring response strips in the same excavation coordinate.

Placeholder for sensitivity figure: parameter-sensitivity heatmap and top- k provenance-overlap bars.

Figure 7: Sensitivity of the component-level geological anomaly index and edge-level provenance to graph-construction settings. The panel reports stability of $I_{shield}^A(t)$, $S_{shield}(t)$ and top- k edge overlap under variations in τ_{edge} , η_{min} and τ_{zone} .

pressure, friction or risk probability. The component-level geological anomaly index is an edge-weighted anomaly summary over a screened relation support; it is not a calibrated failure probability. Response-alignment diagnostics can support consistency with the event, but they should not be presented as causal proof of a unique failure mechanism.

6.3 Implications for spatial data modelling in TBM excavation

The proposed representation treats TBM excavation as a moving spatial relation process. Rock voxels, TBM surface nodes, candidate rock-TBM relations, edge attributes and edge-level provenance form a graph-structured description of the excavation scene. This representation can support spatial queries such as which anomalous voxels are related to a given shield component, which candidate edges contribute most to a component-level index, and how the support of rock-TBM relations changes across excavation steps.

The same relation layer can be linked to future prediction modules without being absorbed into a black-box model. A monitoring-based predictor could use $I_c(t)$, $\Delta I_c(t)$, $A_c(t)$ or edge-provenance summaries as structured inputs, while the graph would retain the entity and edge records needed for spatial interpretation. This separation is important because operational prediction and spatial provenance answer different questions.

Table 6: Interpretation capacity of the proposed graph and alternative geological summaries.

Summary type		Component label	Normal compatibility	Edge lifecycle	Edge provenance	Interpretation
Global anomaly	active-zone	No	No	Partial	No	We need
Pooled TBM anomaly		No	Yes	Yes	Partial	The
No-normal graph	component	Yes	No	Yes	Yes	and
No-state graph	component	Yes	Yes	No	Yes	Con
Full proposed graph		Yes	Yes	Yes	Yes	may
						ble
						Con
						may
						rep
						vox
						An
						spa
						spe
						nen

6.4 Limitations and future work

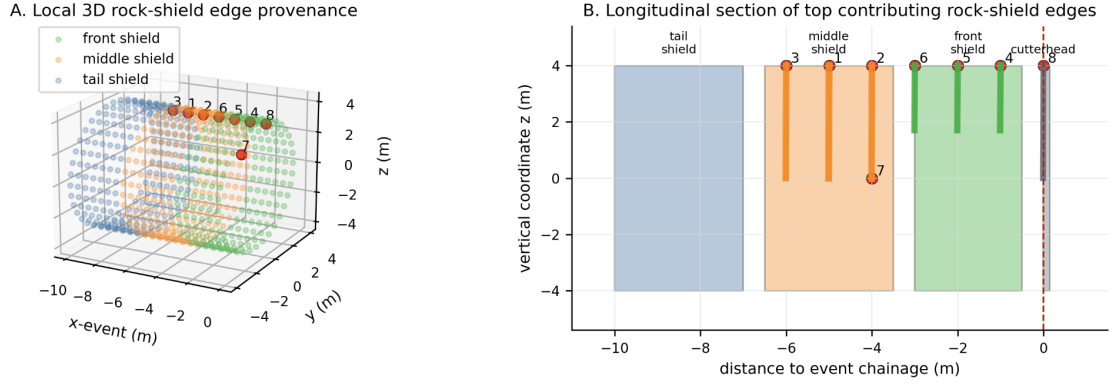
The main limitation is the event-centred nature of the evaluation. A confirmed shield-jamming event provides a strong engineering anchor, but one event cannot establish universal prediction ability. Future work should test the same representation on additional shield-jamming, non-jamming and mixed-anomaly intervals. The second limitation is the treatment of TSP-derived anomaly scores as deterministic voxel attributes. Future work should propagate TSP uncertainty through q_i , test alternative geological anomaly scores and examine how uncertainty affects top-edge provenance. The third limitation is parameter dependence in the graph-construction stage. The sensitivity design proposed in this paper should be expanded into a multi-case robustness analysis once more events become available.

7 Conclusion

This study proposed a dynamic rock–TBM spatial relation graph for component-level geological anomaly interpretation during TBM excavation. The graph represents TSP-derived rock voxels and component-labelled TBM surface nodes as heterogeneous spatial entities, constructs candidate rock–TBM relations under active-zone, distance, surface-normal compatibility and excavation-state constraints, and computes component-level geological anomaly indices from the resulting edge-weighted relations.

The revised evaluation is framed as event-centred representation validation using a confirmed shield-jamming event in the Boshulaling Tunnel. The graph places TSP-derived anomalies, TBM components and monitoring responses in a common excavation coordinate system; reconstructs how low-velocity rock voxels become spatially associated with shield components during event development; compares component-level indices with alternative summaries and ablated graph settings; and traces the interpretation back to specific rock voxels, TBM surface nodes and candidate edges.

The primary value of the proposed graph is to make spatial relations between anomalous geological units and moving TBM components explicit, component-level and traceable. It should be understood as a queryable spatial relation representation for event interpretation and future



C. Top edge-level provenance records near the stuck state

Rank	Component	Rel. chainage	Distance	Normal	q_i	$w_{ij}(t)$	Contribution
1	middle shield	-5.000	0.051	0.982	0.891	0.958	0.853
2	middle shield	-4.000	0.051	0.982	0.829	0.958	0.794
3	middle shield	-6.000	0.051	0.982	0.820	0.958	0.785
4	front shield	-1.000	0.051	0.982	0.806	0.958	0.772
5	front shield	-2.000	0.051	0.982	0.783	0.958	0.750
6	front shield	-3.000	0.051	0.982	0.768	0.958	0.735
7	middle shield	-4.000	0.051	0.982	0.734	0.958	0.703
8	cutterhead	0.000	0.010	1.000	0.694	0.995	0.691

Figure 8: Edge-level provenance of the component-level geological anomaly index near the jammed state. Top contributing edges retain rock voxel identity, TBM surface-node identity, component label, relation geometry and geological anomaly score.

model integration, rather than as a standalone calibrated failure-prediction or contact-force estimation model.

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Disclosure statement

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Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The shareable materials include experiment configurations, graph-construction scripts, figure scripts, derived component-index summaries, alternative-representation diagnostics, sensitivity summaries and edge-level provenance tables. These materials are available from the corresponding author upon reasonable request, subject

Table 7: Recommended edge-level provenance fields for the top candidate relations.

Field	Symbol or example	Purpose
Rock voxel ID	i	Identifies the geological spatial support
TBM surface-node ID	j	Identifies the machine-side spatial support
Component label	ρ_j	Links the edge to cutter-head/front/middle/tail shield
Relative chainage	$x_i - x_{event}$	Locates the rock voxel relative to the confirmed event
Distance	$d_{ij}(t)$	Reports metric proximity
Normal compatibility	$\kappa_{ij}(t)$	Reports facing-geometry plausibility
Geometry weight	$w_{ij}^{rm}(t)$	Reports edge relation weight
Anomaly score	q_i	Reports geological contribution
Weighted contribution	$s_{ij}(t) = w_{ij}^{rm}(t)q_i$	Ranks edge-level provenance

to project data-use permissions. To support reproducibility without disclosing site data, the authors can provide a synthetic demonstration package containing a small voxel field, component-labelled TBM surface nodes, candidate edge lists, component-level geological anomaly-index scripts, and edge-level provenance examples.

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Table 8: Template for supplementary component–response association diagnostics. Correlations are exploratory and should not be interpreted as calibrated prediction performance.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL $h = 1$	Cutterhead					
BSLL $h = 1$	Front shield					
BSLL $h = 1$	Middle shield					
BSLL $h = 1$	Tail shield					
BSLL $h = 3$	Cutterhead					
BSLL $h = 3$	Front shield					
BSLL $h = 3$	Middle shield					
BSLL $h = 3$	Tail shield					

A Supplementary diagnostics and robustness checks

Table 8 should retain the full unadjusted component–response Spearman matrix only as an exploratory supplementary diagnostic. The main text should report a smaller response-alignment summary tied to the shield-jamming event and should avoid using unadjusted correlations as primary evidence.

Suggested additional supplementary diagnostics include: (i) null windows outside the event interval, (ii) randomised component labels, (iii) shuffled anomaly scores within the active zone, and (iv) uncertainty perturbations of q_i .